

ED BUELER

PROFESSOR OF MATHEMATICS (APPLIED)

Dept. Mathematics & Statistics
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CV version: October 8, 2026

ACADEMIC POSITIONS AND EDUCATION

Professor, DMS, UAF (2017–current).
Associate (2005–2017) and Assistant (1998–2005) Professor, DMS, UAF.
Postdoctoral Fellow, *Year in Stochastic Analysis*, Mathematical Sciences Research Institute, Berkeley, August 1997–May 1998.
Ph. D. Mathematics, Cornell University, Ithaca NY, 1997. (advisor: Leonard Gross; thesis: *The heat kernel weighted Hodge Laplacian on noncompact manifolds*)
M. S. Mathematics, Cornell University, Ithaca NY, 1994.
B. S. Mathematics with Honors, Minor in Physics, Minor in Electrical Engineering, California State University, Chico CA, 1991.

HONORS AND AWARDS

Nelder Visiting Fellowship, Imperial College London, 2026.
Department Outstanding Teacher 2023–2024.
Student nominations for Robert Piacenza Excellence in Teaching Award: 2022, 2025.
Geophysical Institute 2015 Best Faculty Paper Award.
Faculty Advising Award for Outstanding Undergraduate Advising, 2003–2004.
Honorary Faculty Certificate of Appreciation for support of a student-athlete, 2004.
Bonus for Extraordinary Performance, CSEM, 2003.

BOOK AND BOOK CHAPTER

E. Bueler, *PETSc for Partial Differential Equations: Numerical Solutions in C and Python*, SIAM Press 2021. Supporting codes at github.com/bueler/p4pdes.
E. Bueler, “Chapter 8. Numerical modeling of ice sheets, streams, and shelves,” in *Glaciers and Ice Sheets in the Climate System: The Karthaus Summer School Lecture Notes*, A. Fowler and F. Ng editors, Springer 2021.

PUBLICATIONS (PEER-REVIEWED)

E. Bueler, 2026. *Pointwise a posteriori error bounds for bilateral classical obstacle problems*, in preparation
S. Fochesatto and E. Bueler, 2026. *VIAMR: Adaptive mesh refinement and adaptation for variational inequalities and obstacle problems*, in preparation
E. Bueler, 2025. *Surface elevation errors in finite element Stokes models for glacier evolution*, preprint [arxiv.org:2408.06470](https://arxiv.org/abs/2408.06470), submitted.
30. E. Bueler and P. E. Farrell, 2024. *A full approximation scheme multilevel method for non-linear variational inequalities*, SIAM J. Sci. Comput. 46 (4).

29. E. Bueler, 2023. *Performance analysis of high-resolution ice sheet simulations*, J. Glaciol., 69 (276), 930–935.
28. E. Bueler, 2021. *Conservation laws for free-boundary fluid layers*, SIAM J. Appl. Math., 81 (5), 2007–2032.
27. D. J. Brinkerhoff, C. R. Meyer, E. Bueler, and M. Truffer, 2016. *Inversion of a glacier hydrology model*, Ann. Glaciol. 57 (72), 84–95.
26. E. Bueler, 2016. *Stable finite volume element schemes for the shallow-ice approximation*, J. Glaciol. 62 (232), 230–242.
25. E. Bueler and W. van Pelt, 2015. *Mass-conserving subglacial hydrology in the Parallel Ice Sheet Model*, Geoscientific Model Development 8 (6), 1613–1635.
24. E. Bueler, 2014. *An exact solution for a steady, flow-line marine ice sheet*, J. Glaciol. 60 (224), 1117–1125.
23. E. Bueler, 2014. *Correspondence: Extending the lumped subglacial-englacial hydrology model of Bartholomaus and others (2011)*, J. Glaciol. 60 (222), 80–810.
22. S. Nowicki and 30 others including E. Bueler, 2013. *Insights into spatial sensitivities of ice mass response to environmental change from the SeaRISE ice sheet modeling project II: Greenland*, J. Geophys. Res. (Earth Surface) 118 (2), 1025–1044.
21. S. Nowicki and 30 others including E. Bueler, 2013. *Insights into spatial sensitivities of ice mass response to environmental change from the SeaRISE ice sheet modeling project I. Antarctica*, J. Geophys. Res. (Earth Surface), 118 (2), 1002–1024.
20. G. Juvet, E. Bueler, C. Gräser, and R. Kornhuber, 2013. *A nonsmooth Newton multigrid method for a hybrid, shallow model of marine ice sheets*, AMS Contemporary Mathematics (SCA 2012) 586, 197–205.
19. G. Juvet and E. Bueler, 2012. *Steady, shallow ice sheets as obstacle problems: well-posedness and finite element approximation*, SIAM J. Appl. Math. 72 (4), 1292–1314.
18. A. Aschwanden, E. Bueler, C. Khroulev, and H. Blatter, 2012. *An enthalpy formulation for glaciers and ice sheets*, J. Glaciol. 58 (209), 441–457.
17. F. Pattyn, C. Schoof, and 16 others including E. Bueler, 2012. *Results of the Marine Ice Sheet Model Intercomparison Project, MISIP*, The Cryosphere 6, 573–588.
16. M. A. Martin, R. Winkelmann, M. Haseloff, T. Albrecht, E. Bueler, C. Khroulev, and A. Levermann, 2011. *The Potsdam Parallel Ice Sheet Model (PISM-PIK)–Part 2: Dynamic equilibrium simulation of the Antarctic ice sheet*, The Cryosphere 5, 727–740.
15. R. Winkelmann, M. A. Martin, M. Haseloff, T. Albrecht, E. Bueler, C. Khroulev, and A. Levermann, 2011. *The Potsdam Parallel Ice Sheet Model (PISM-PIK)–Part 1: Model description*, The Cryosphere 5, 715–726.
14. G. Juvet, J. Rappaz, E. Bueler, and H. Blatter, 2011. *Existence and stability of steady state solutions of the shallow ice sheet equation by an energy minimization approach*, J. Glaciol. 57 (202), 345–354.
13. R. Calov, R. Greve, A. Abe-Ouchi, E. Bueler, P. Huybrechts, J. V. Johnson, F. Pattyn, D. Pollard, C. Ritz, F. Saito, and L. Tarasov, 2010. *Results from the Ice-Sheet Model Intercomparison Project—Heinrich Event IntercOmparison (ISMIP–HEINO)*, J. Glaciol. 56 (197), 371–383.
12. E. Bueler and J. Brown, 2009. *Shallow shelf approximation as a “sliding law” in a thermo-mechanically coupled ice sheet model*, J. Geophys. Res. (Earth Surface) 114, F03008.
11. E. A. Butcher, M. Sari, E. Bueler, and T. Carlson, 2009. *Magnus’ expansion for time-periodic systems: parameter-dependent approximations*, Communications in Nonlinear Sciences and Numerical Simulation 14, 4226–4245.

10. E. A. Butcher, O. A. Bobrenkov, E. Bueler, and P. Nindujarla, 2009. *Analysis of milling stability by the Chebyshev collocation method: Algorithm and optimal stable immersion levels*, Journal of Computational and Nonlinear Dynamics (ASME) 4 (3), 031003.
9. V. Deshmukh, E. A. Butcher, and E. Bueler, 2008. *Dimensional reduction of nonlinear delay differential equations with periodic coefficients using Chebyshev spectral collocation*, Nonlinear Dynamics 52, 137–149.
8. E. Bueler, 2007. *Error bounds for approximate eigenvalues of periodic-coefficient linear delay differential equations*, SIAM J. Num. Analysis 45 (6), 2510–2536.
7. E. Bueler, J. Brown, and C. Lingle, 2007. *Exact solutions to the thermomechanically coupled shallow ice approximation: effective tools for verification*, J. Glaciol. 53 (182), 499–516.
6. E. Bueler, C. S. Lingle, and J. A. Brown, 2007. *Fast computation of a viscoelastic deformable earth model for ice flow simulations*, Ann. Glaciol. 46, 97–105.
5. E. Bueler, C. S. Lingle, J. A. Kallen-Brown, D. N. Covey, and Latrice N. Bowman, 2005. *Exact solutions and the verification of numerical models for isothermal ice sheets*, J. Glaciol. 51 (173), 291–306.
4. E. A. Butcher, H. Ma, E. Bueler, V. Averina, and Z. Szabo, 2004. *Stability of time-periodic delay-differential equations via Chebyshev polynomials*, International Journal on Numerical Methods in Engineering 59 (7), 895–922.
3. E. A. Butcher, H. Ma, and E. Bueler, 2003. *Chebyshev expansion of linear dynamic systems with time delay and periodic coefficients under control excitations*, Journal of Dynamic Systems, Measurement and Control (ASME) 125, 236–243.
2. E. Bueler and I. Prokhorenkov, 2002. *Hodge theory and cohomology with compact supports*, Soochow Journal of Mathematics 28 (1), 33–55.
1. E. Bueler, 1999. *The heat kernel weighted Hodge Laplacian on noncompact manifolds*, Transactions of the American Mathematical Society 351, 683–713.

EDITOR-REVIEWED PUBLICATIONS

6. W. Lipscomb, R. Bindshadler, E. Bueler, D. Holland, J. Johnson, and S. Price, 2009. *Building a Next-Generation Community Ice Sheet Model*, Eos Transactions, AGU, 90 (3), 23.
5. E. Bueler, *Lessons from the short history of ice sheet model intercomparison*, The Cryosphere Discussions 2, 1–14, 2008.
4. E. A. Butcher, V. Deshmukh, and E. Bueler, *Center manifold reduction of periodic delay differential systems*, Proceedings of the ASME IDETC/CIE, 2007.
3. E. A. Butcher, P. Nindujarla and E. Bueler, *Stability of up- and down-milling using a Chebyshev collocation method*, Proceedings of ASME IDETC/CIE, 2005.
2. V. Averina and four others, *Effect of delay on engine air-to-fuel ratio control*, Proceedings of the IEEE Conference on Control Applications, Toronto, 2005.
1. E. A. Butcher and four others, *Stability analysis of parametrically excited systems with time-delay*, Proceedings of ASME DETC, 2003.

TECHNICAL REPORTS

8. E. Bueler, *The full approximation storage multigrid scheme: A 1D finite element example*, arxiv:2101.05408, 2021.
7. E. Bueler, *An exact solution to the temperature equation in a column of ice and bedrock*, arXiv:0710.1314, 2007.
6. E. Bueler, *Chebyshev collocation for linear, periodic ordinary and delay differential equations: a posteriori estimates*, arXiv:math.NA/0409464 (2004).

5. E. Bueler, *Construction of steady state solutions for isothermal shallow ice sheets*, UAF DMS Tech. Rep. 03–02 (2003).
4. E. Bueler and E. A. Butcher, *Stability of periodic linear delay–differential equations and the Chebyshev approximation of fundamental solutions*, UAF DMS Tech. 02–03 (2002).
3. E. Bueler, *Numerical approximation of a two dimensional thermomechanical model for ice flow*, UAF DMS Tech. Rep. 02–02 (2002).
2. E. Bueler, *Dirac operators as “annihilation operators” on Riemannian manifolds*, (2001).
1. E. Bueler, *Number operators for Riemannian manifolds*, [arXiv:math-ph/0104022](https://arxiv.org/abs/math-ph/0104022) (2000).

SOFTWARE FOR SCIENTIFIC COMPUTING

- **Parallel Ice Sheet Model (PISM):**

github.com/pism, pism.io

PISM is an open-source, numerical fluid simulation code for ice sheets and glaciers based on the modern, parallel HPC libraries [PETSc](https://petsc.org) and MPI. It contains more than 100,000 lines of C++ and Python. Most of the source code and all of the user support and documentation is through our group at UAF, but some parts were co-developed with the Potsdam Institute for Climate Impact Research, Germany. The primary authors are: C. Khroulev (lead programmer/author), A. Aschwanden (current Principal Investigator), E. Bueler (programmer/author and first Principal Investigator), J. Brown (original programmer/author), D. Maxwell (inversion codes), and T. Albrecht (ice shelf physics). PISM has been funded by twelve major NASA and NSF grants, totaling more than \$5 million. Since 2007 there have been at least 221 peer-reviewed journal articles which apply PISM or describe its design (pism.io/publications/). Of these papers I am a (co-)author of 10: 6, 7, 12, 13, 15, 16, 18, 21, 22, 25 in publication list.

- **Example programs for *PETSc* for *Partial Differential Equations*:**

github.com/bueler/p4pdes

A collection of programs in support of my book. Each code uses [PETSc](https://petsc.org) to solve a PDE or related problem. The C codes call PETSc directly while the Python codes apply the [Firedrake](https://firedrakeproject.org) finite element library, which applies PETSc as a solver library.

- **Teaching glacier and ice sheet modeling:**

1. github.com/bueler/mccarthy is a suite of introductory codes in Matlab and Python, plus notes and slides, from my lectures at the International Summer School in Glaciology, McCarthy, Alaska and the Summer School on Ice Sheets and Glaciers in the Climate System, Karthaus, Italy; see also my book chapter.
2. github.com/bueler/stokes-ice-tutorial uses [Firedrake](https://firedrakeproject.org) (finite elements) for the glaciological Stokes equations.

- **VIAMR = Adaptive mesh refinement on variational inequalities:**

github.com/StefanoFochesatto/viamr

Joint with Stefano Fochesatto, using [Firedrake](https://firedrakeproject.org) and Python.

- **DDEC = stability charts of linear periodic delay differential equations:**

bueler.github.io/ddes/DDEcharts.htm

A suite of *Matlab* programs used in publications 8, 9, 10.

FUNDED RESEARCH GRANTS

E. Bueler, Faculty Development (UNAC) Travel Award, 2023, \$2k.

E. Bueler (PI), M. Fahnestock (Co-I), A. Aschwanden (Co-I), *Understanding Measured Variability in the Greenland Ice Sheet Using the Parallel Ice Sheet Model (PISM)*, NASA Modeling Analysis and Prediction Program grant #NNX13AM16G, June 2013–June 2018, \$700k.

E. Bueler (PI), R. Hock (Co-I), D. Maxwell (Co-I), and M. Truffer (Co-I), *A high resolution Parallel Ice Sheet Model including fast, sliding flow: advanced development and application*, NASA Modeling Analysis and Prediction Program grant #NNX09AJ38G, June 2009–June 2013, \$997k.

C. Lingle (PI), D. Covey (Co-I), and E. Bueler (Co-I), *Ice Sheet Modeling: a component of NSF grant PRISM: Polar Radar for Ice Sheet Measurements* NASA Cryospheric Sciences Program grant #NAG5-11371, October 2001–September 2006, \$436k.

E. A. Butcher (PI) and E. Bueler (Co-PI), *Symbolic Stability and Bifurcation Analysis of Time-Periodic Differential-Delay Equations: Applications to High-Speed Machining Models*, NSF Civil and Mechanical Systems Directorate for Engineering #0114500, September 2001–August 2004, \$205k, plus \$5k supplemental for one REU student.

SUPERVISOR/MENTOR OF

- Constantine Khroulev, GI Research Professional, 2009–2018.
- Dr. Andreas Aschwanden, ARSC and GI Postdoctoral Fellow, 2009–2013.

ADVISING OF GRADUATE RESEARCH

- Nicholas Harrison, M.S. Mathematics, Project: *Well-posedness and numerical approximation of a coupled predator-prey system*, UAF, May 2026.
- Emmanuel Azorko, M.S. Mathematics, Project: *Finite element treatment of the Signorini problem in linear elasticity: The frictionless beam case*, UAF, May 2026.
- Stefan Fochesatto, M.S. Mathematics, Project: *Adaptive mesh refinement for variational inequalities* ([arxiv:2502.14206](https://arxiv.org/abs/2502.14206)), UAF, December 2024.
- Wesley Voley, M.S. Mathematics, Thesis: *Navier-Stokes equations: An overview of existence and regularity theorems*, UAF, August 2021.
- Max Heldman, M.S. Mathematics, Project: *Toward an optimal solver for the obstacle problem*, UAF, May 2018.
- Lyman Gillispie, M.S. Mathematics, Thesis: *A temperature-only formulation for ice sheets*, UAF, May 2014.
- William Mitchell, M.S. Mathematics, Thesis: *Exact and numerical solutions for Stokes flow in glaciers*, UAF, August 2012.
- Daniella DellaGiustina (co-advisee), M.S. Computational Physics, Thesis: *Regional modeling of Greenland's outlet glaciers with PISM*, UAF, December 2011.
- Jacob Stroh, M.S. Mathematics, Thesis: *Non-normality in scalar delay differential equations*, UAF, December 2006.
- Jed A. Kallen-Brown, M.S. Mathematics, Project: *Multi-modal ice sheet dynamics: theory and implementation*, UAF, August 2006.
- Timothy Carlson, M.S. Mathematics, Thesis: *Magnus' expansion as an approximation tool for ODEs*, UAF, May 2005.
- Latrice N. Bowman, M.S. Mathematics, Project: *Stability and accuracy of numerical finite difference methods applied to two dimensional isothermal ice flow*, UAF, November 2002.
- Viktoria A. Averina, M.S. Mathematics, Thesis: *Symbolic stability of delay differential equations*, UAF, August 2002.

- Mikhail Korotiaev, M.S. Mathematics, Thesis: *Critical points of the heat kernel on a compact semisimple Lie group*, UAF, August 2002.
- Liane Hansen, M.S. Mathematics, Project: *Numerical solution of a weighted Hodge Laplacian*, UAF DMS, May 1999.

ADVISING OF UNDERGRADUATE RESEARCH

- Sarah Williamson, Alaska Space Grant Program Undergraduate Fellowship, Spring 2017.
- Jeremiah Harrington, Honors Thesis in Mathematics, May 2013.
- Benjamin Sperisen, ARSC undergraduate Intern, Summer 2008.
- Nathan Shemonski, ARSC undergraduate Intern, Summer 2007.
- Benjamin White, NSF REU, Summer 2004.

COURSES TAUGHT AT UAF (S=SPRING, F=FALL SEMESTER)

Undergraduate.

Math 251 Calculus I (*13 instances, through S20*)
 Math 252 Calculus II (*9 instances, through S26*)
 Math 253 Calculus III (F98, F02, F15, S18, S23)
 Math 302 Differential Equations (S00, F00, S09, S19, F23)
 Math 314 Linear Algebra (S07, S22)
 Math 401 Introduction to Real Analysis (F13, F25)
 Math 404 Topology (F16)
 Math 412 Differential Geometry (S99, S03)
 Math 421 Applied Analysis (F01, F04, F07, F11)
 Math 422 Introduction to Complex Analysis (S08, S16)
 Math 426 Numerical Analysis (*13 instances, through F24*)
 Math 497 Introduction to Dynamics & Chaos (independent study S26)

Graduate.

Math 611/612 Mathematical Physics I (F05) & II (S06)
 Math 614 Numerical Linear Algebra (*new course; 8 instances, through F25*)
 Math 615 Numerical Analysis of Diff. Eqns. (*10 instances, through S25*)
 Math 617 Functional Analysis (S20, S24, S26)
 Math 641 Real Analysis (F00, F01, F19)
 Math 661 Optimization (F16, F18, F22, F24)
 seminars: finite elements (F04, F18, S24), random walks (S01), differential forms (F01), iterative methods (F03), PETSc (S16), machine learning (S22), scalable algorithms (S23), fluids & solids (S25)

SIGNIFICANT PROFESSIONAL ACTIVITIES

- Lecturer on *Numerical modelling of ice sheets and glaciers*, International Summer School in Glaciology, McCarthy, Alaska (2010, 2012, 2014, 2016, 2018, 2022, 2024, 2026), and at the Summer School on Ice Sheets and Glaciers in the Climate System, Karthaus, Italy (2009, 2010, 2012, 2014).
- Contributor to the *Sea-level Response to Ice Sheet Evolution (SeaRISE)* assessment, a NASA-funded international effort to estimate ice sheet contributions to sea level.

- Workshop/minisymposium co-organizer: SIAM Conference on Mathematical and Computational Issues in the Geosciences, Santa Fe 2007; Building a Next-Generation Community Ice Sheet Model, Los Alamos 2008; European PISM Workshop, Max Planck Institute for Meteorology, Hamburg 2012.
- Reviewer of articles for more than a dozen journals in applied mathematics, numerical analysis, geosciences, and glaciology. About four per year.
- Proposal/panel reviewer: DOE (2011,2014,2017), NASA (2012,2016), NSF (2001,2006,2009,2021), Netherlands NSF (2007), U.K. NERC (2010), RC of Norway (2024,2025), ARSC (2010), Austrian FWF (2019), SIAM Geosciences prizes (2018).
- Reviews: 14 summaries for *Mathematical Reviews* from 1998 to 2005; numerical analysis texts for Addison-Wesley (2000, 2001).
- Member of Society for Industrial and Applied Mathematics (SIAM) and the International Glaciological Society (IGS)